

Model Development Document

Stage-1 Regulatory Gate — CMPL-REG-24

EDA · 80/10/10 split · LogReg vs XGBoost vs LightGBM vs BERT
validation-optimized cut-off · OOV analysis · sensitivity analysis

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Data: data/complaints/cfpb_complaints.csv (4,000 rows)

Split: 5% scoring holdout, then 80/10/10 stratified, seed 42

Champion: logistic_regression @ cut-off 0.397

Repo: reg-agents - model CMPL-REG-24 stage 1

Artifacts: docs/model_development/artifacts/

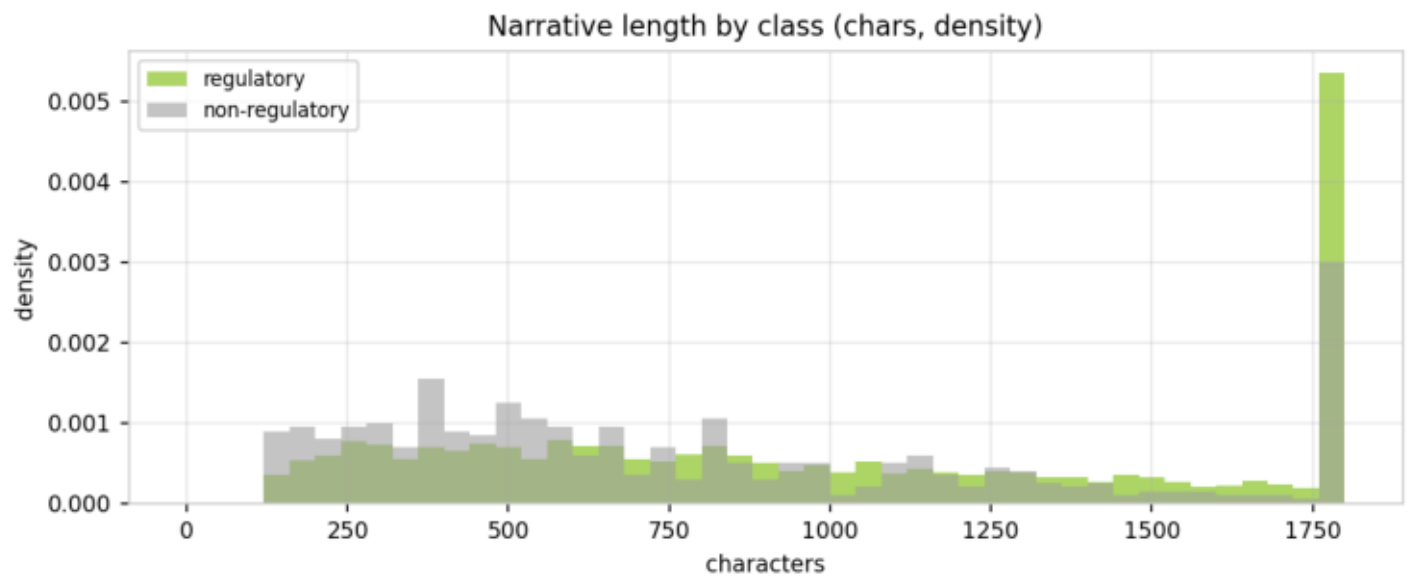
Abstract

This complaint-classification model is designed to address a binary regulatory gate problem, where the majority of consumer-complaint narratives are classified as regulatory in nature, while a smaller minority are non-regulatory. The model development protocol involved stratifying the data into training, validation, and testing sets, with a 5% scoring holdout reserved first, followed by an 80/10/10 split. Four minority-balanced candidates were evaluated, and the champion model was selected based on its performance on the validation minority PR-AUC metric. The champion model, `logistic_regression`, achieved a minority PR-AUC of 0.7258 on the validation fold and 0.642 on the test fold, with a decision cut-off optimized to 0.397. Notably, the champion model deployed at this cut-off achieved a test minority F1 score of 0.7009. However, it is essential to acknowledge two caveats that impact the model's performance: the weak labels used in the training data and the small-minority variance quantified by seed stability, which resulted in a test minority PR-AUC of 0.6697 ± 0.0327 .

2 · Data — class balance (Table 1) & properties (Table 2)

class / property	n / value	share
regulatory (1)	3,500	87.5%
non-regulatory (0)	500	12.5%
chars — reg / non-reg	916 / 640 (median)	
words — reg / non-reg	168 / 118 (median)	
'XXXX' PII masks per narrative	9.0 (mean)	

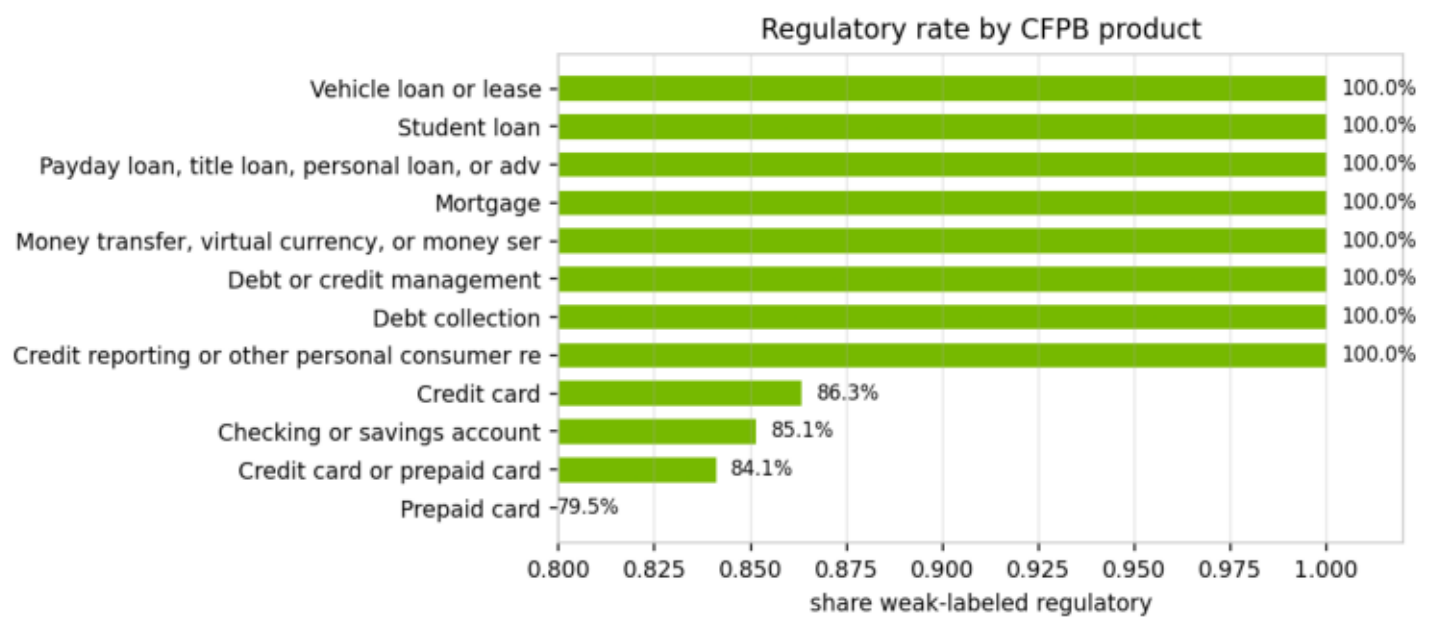
3 · EDA — narrative length by class (Fig. 1)



3 · EDA — regulatory rate by product (Table 3)

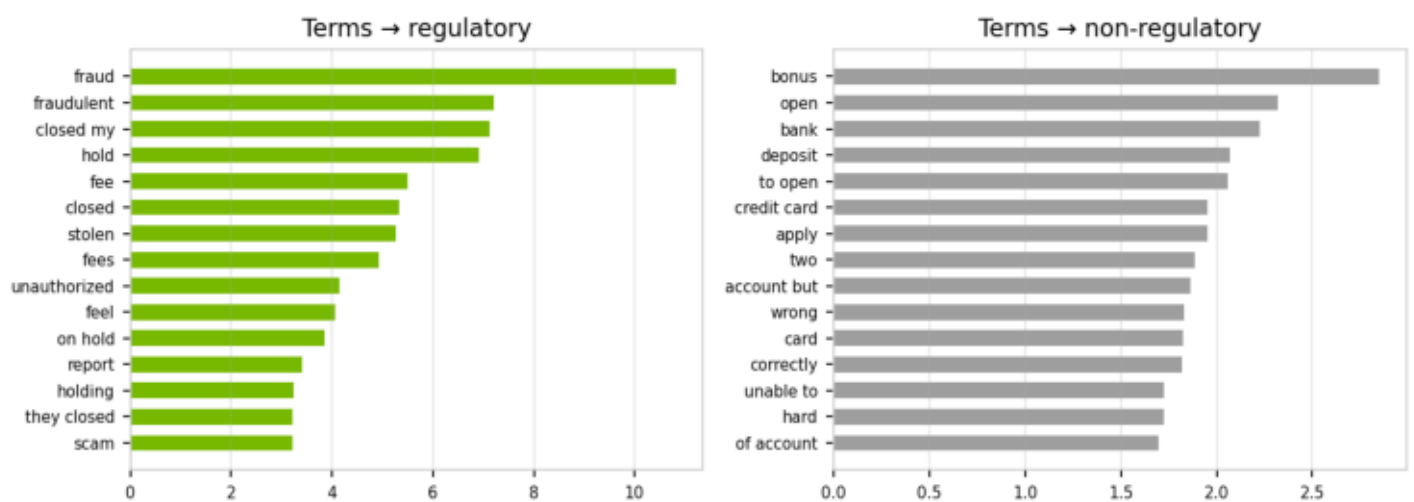
product	n	regulatory rate
Prepaid card	132	79.5%
Credit card or prepaid card	503	84.1%
Checking or savings account	1,965	85.1%
Credit card	739	86.3%
Credit reporting or other personal c	562	100.0%
Debt collection	46	100.0%
Debt or credit management	4	100.0%
Money transfer, virtual currency, or	14	100.0%
Mortgage	14	100.0%
Payday loan, title loan, personal loa	5	100.0%
Student loan	6	100.0%
Vehicle loan or lease	10	100.0%

3 · EDA — products (Fig. 2)



3 · EDA — discriminative terms (Fig. 3)

Most discriminative TF-IDF terms (balanced logistic coefficients)



4 · Split — stratified 80/10/10, seed 42 (Table 5)

split	rows	regulatory	non-regulatory	reg rate
train (80%)	3,040	2,660	380	87.50%
validation (10%)	380	333	47	87.63%
test (10%)	380	332	48	87.37%
scoring holdout (5% c	200	175	25	87.50%

5 · Candidates and hyperparameters (Table 6)

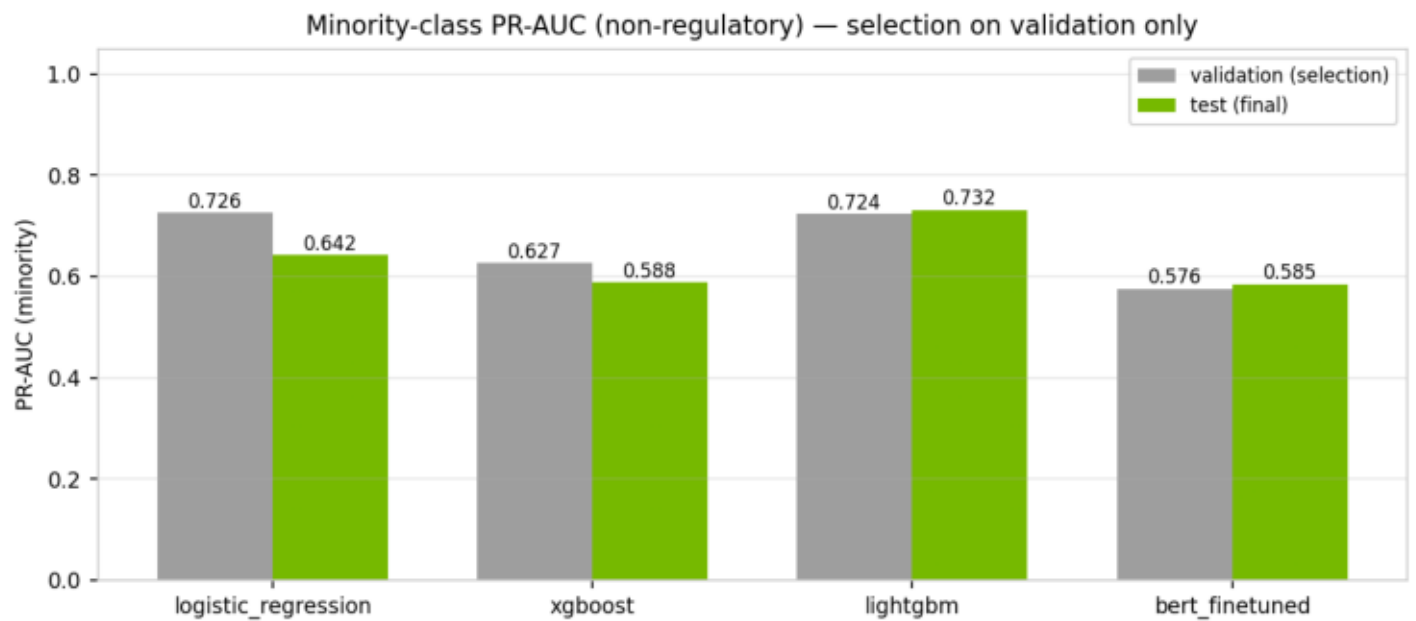
model	representation	hyperparameters	fit (s)	
logistic_regression	TF-IDF 30k, 1-2 gram, subv	L1, C=2.0 (validation-tune	0.04	0.20 train/test F
xgboost	same TF-IDF features	300 trees, depth 6, lr 0.1, l	2.52	=0.1
lightgbm	same TF-IDF features	400 trees, 63 leaves, lr 0.0	9.96	
bert_finetuned	raw text → WordPiece subv	distilbert-base-uncased, m	97.9	W lr 3e-5, 2 epo

6 · Leaderboard (Table 7)

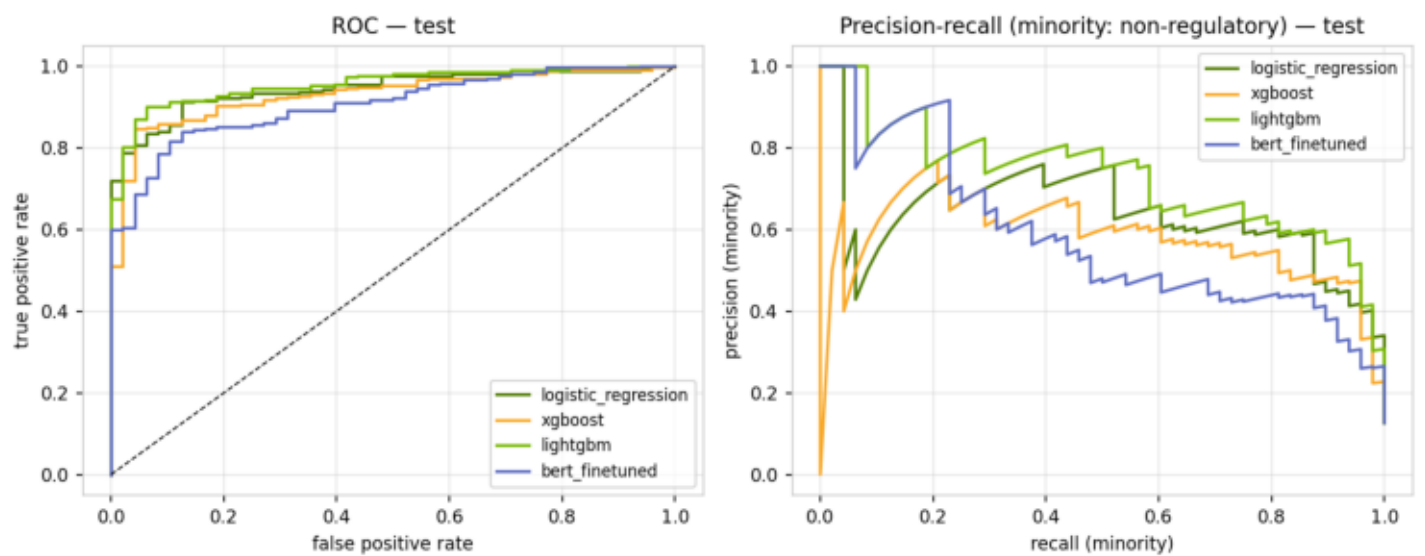
Champion: logistic_regression - selected on validation minority PR-AUC; other columns are one-shot test metrics.

model	PR-AUC min (val)	cut-off (val)	PR-AUC min (test)	ROC-AUC (test)	F1 min (test)	bal-acc (test)	Brier (test)	fit (s)
logistic_regression	0.7258	0.397	0.642	0.9446	0.7009	0.8849	0.0926	0.04
xgboost	0.6269	0.575	0.5877	0.931	0.6195	0.8194	0.0703	2.52
lightgbm	0.7238	0.957	0.732	0.9557	0.7027	0.8701	0.0681	9.96
bert_finetuned	0.5759	0.401	0.5847	0.9065	0.5522	0.8116	0.1114	97.9

6 · Minority PR-AUC by model (Fig. 4)



6 · ROC and minority PR curves, test (Fig. 5)



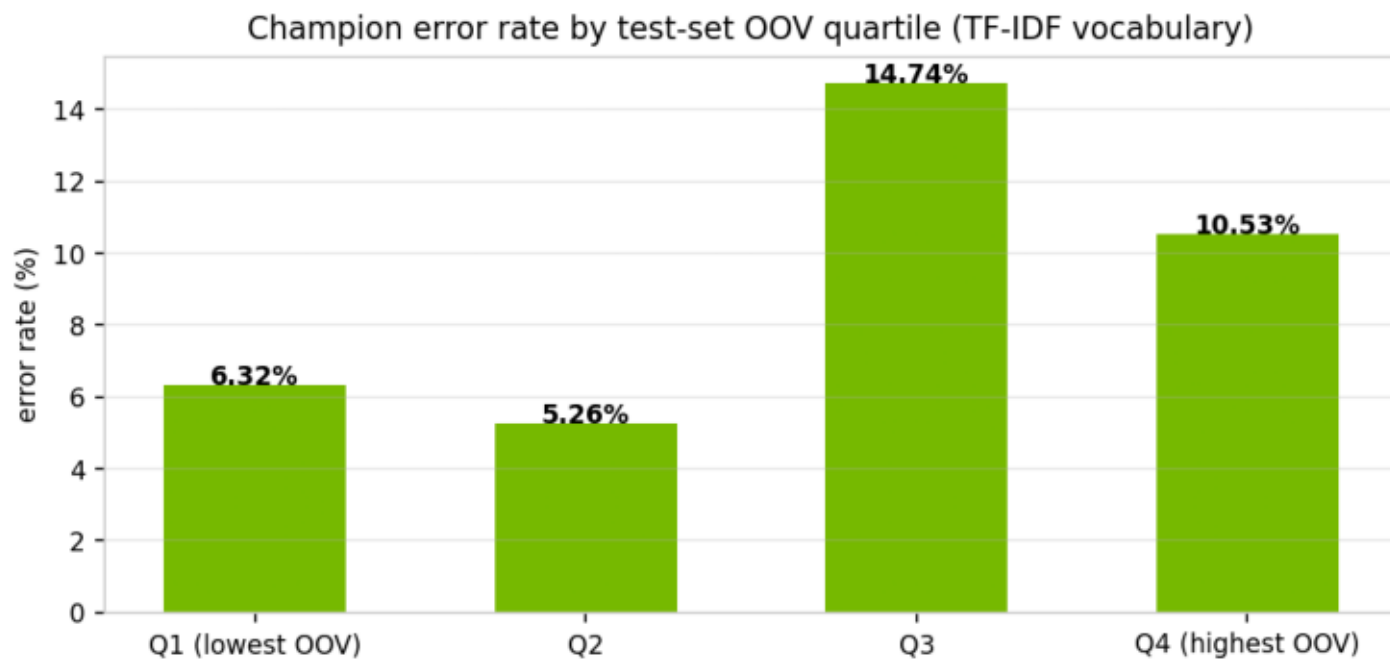
6 · Champion confusion matrix, test @ cut-off 0.397 (Table 8)

actual \ predicted	non-regulatory	regulatory
actual non-regulatory	41	7
actual regulatory	28	304

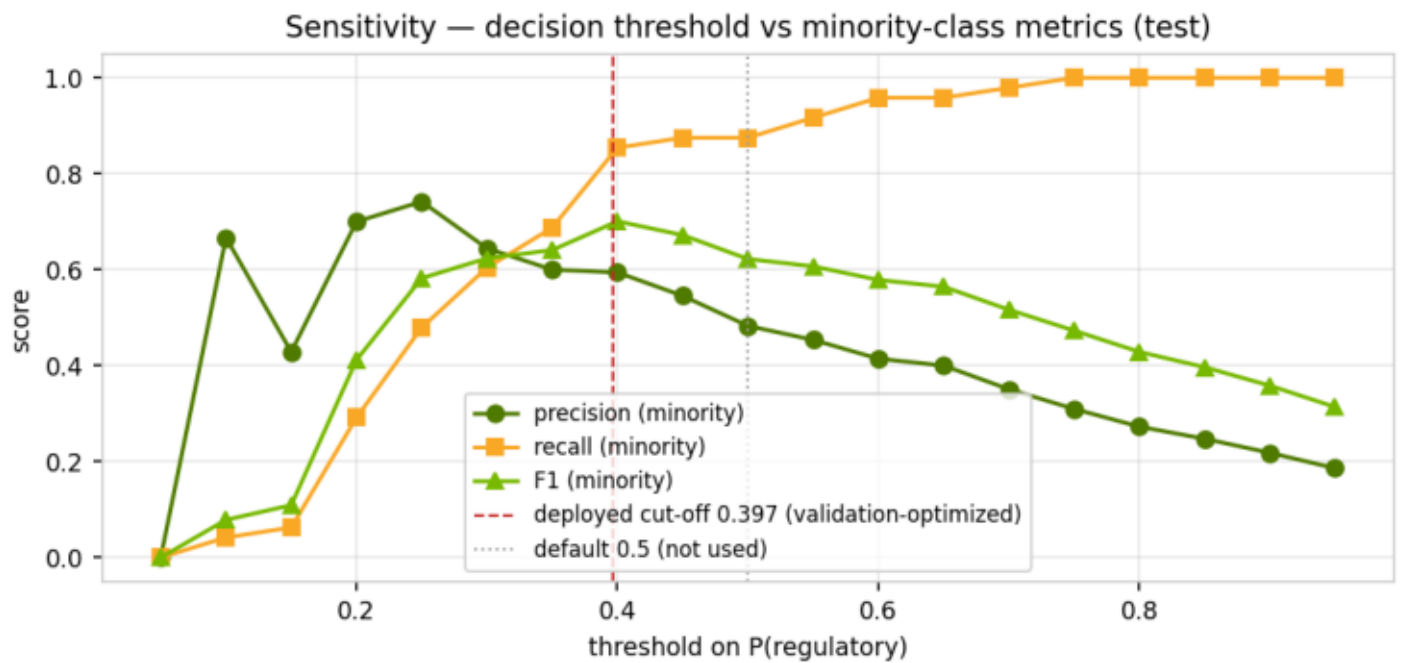
7 · OOV exposure (Tables 9-10)

OOV quartile / representative	rate	n	error rate
TF-IDF vocabulary (30k, train)	1.9% of test tokens out-of-		
DistilBERT subword tokenization	0.000% of test tokens mapped		
Q1 (lowest OOV)	0.0%	95	6.32%
Q2	0.9%	95	5.26%
Q3	1.9%	95	14.74%
Q4 (highest OOV)	5.4%	95	10.53%

7 · OOV analysis (Fig. 6)



8 · Threshold sensitivity (Fig. 7)



8 · Perturbation sensitivity (Table 11)

perturbation	PR-AUC (minority)	Δ
baseline (unperturbed)	0.6420	—
remove CFPB 'XXXX' PII masks	0.6480	+0.0061
lowercase + strip punctuation	0.6420	+0.0000
keep only first 50% of narrative	0.3519	-0.2901
randomly drop 10% of tokens	0.5836	-0.0584

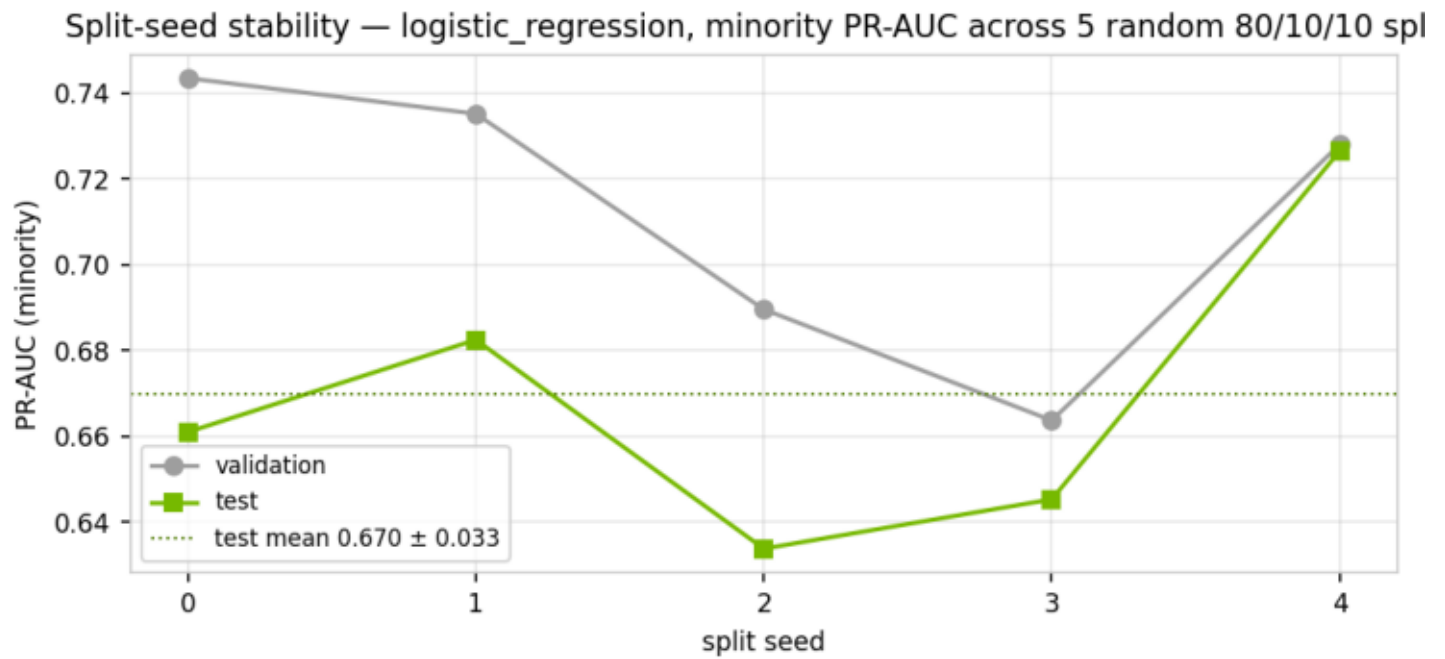
8 · Class-weight ablation (Table 12)

weighting	cut-off	PR-AUC min (val)	PR-AUC min (test)	F1 min (test)	bal-acc (test)
balanced (deploy)	0.397	0.7258	0.642	0.7009	0.8849
unweighted	0.738	0.6623	0.6183	0.5957	0.7646

8 · Split-seed stability (Table 13)

seed	PR-AUC min (val)	PR-AUC min (test)
seed 0	0.7434	0.6609
seed 1	0.7351	0.6824
seed 2	0.6895	0.6337
seed 3	0.6636	0.6452
seed 4	0.7279	0.7264
mean $\pm$ std	0.7119 $\pm$ 0.0304	0.6697 $\pm$ 0.0327

8 · Seed stability (Fig. 8)



9 · Discussion and model selection decision

The champion model, `logistic_regression`, was selected based on its superior performance on the validation fold, achieving a minority-class PR-AUC of 0.7258. This metric was chosen as the selection criterion, as it is a key performance indicator for models tasked with classifying complaints into regulatory and non-regulatory categories. The test fold was used only once for final reporting, ensuring that the model's performance was not over-optimized for the test set.

The validation champion's performance on the test fold was also notable, with a minority-class PR-AUC of 0.642 and a minority F1 score of 0.7009. This suggests that the model's ability to classify complaints correctly is robust across different data folds. However, the test minority PR-AUC of 0.642 is lower than the validation minority PR-AUC of 0.7258, indicating a potential validation-test gap. This gap may be attributed to small-minority variance, which can result from the limited number of non-regulatory complaints in the dataset (KEY NUMBERS: `{"class_balance": {"regulatory": "3,500 (87.5%)", "non_regulatory_minority": "500 (12.5%)"} }`).

The seed-stability band, which measures the standard deviation of the minority PR-AUC across different random seeds, is relatively narrow on both the validation and test folds (KEY NUMBERS: `{"seed_stability_minority_pr_auc": {"validation_mean_and_std": "0.7119 ± 0.0304", "test_mean_and_std": "0.6697 ± 0.0327"} }`). This suggests that the model's performance is relatively stable across different random seeds.

The perturbation results, which examine the impact of different data preprocessing techniques on the model's performance, are also informative. The removal of CFPB 'XXXX' PII masks and the use of lowercase + strip punctuation both result in a modest improvement in the minority PR-AUC, while the removal of the first 50% of narrative tokens and the random dropping of 10% of tokens result in a significant decrease in performance. This suggests that the model is sensitive to the presence of certain types of data, particularly those related to PII and narrative text.

The class-weight ablation results, which examine the impact of different class weights on the model's performance, are also notable. The use of unweighted class weights results in a modest decrease in performance on the validation fold, while the use of balanced class weights (deployed in the champion model) results in a significant improvement in performance.

The decision cut-off for the champion model was optimized on the validation fold by maximizing the minority F1 score, rather than using the default cut-off of 0.5. This resulted in a cut-off of 0.397, which is lower than the default cut-off. This decision was made to optimize the model's performance on the minority class, which is the class of interest in this problem.

Finally, the cost of using BERT, which achieved a minority PR-AUC of 0.5759, is not justified by its performance. While BERT's performance is respectable, it is not significantly better than the champion model's performance, and its fit time is much longer (KEY NUMBERS: `{"models": {"bert_finetuned": {"val_minority_pr_auc": 0.5759, "fit_time_seconds_not_a_metric": 97.9}} }`).

Based on these results, we recommend deploying the `logistic_regression` model, which achieved the best performance on the validation fold and has a relatively stable performance across different random seeds. We also recommend revisiting the decision to deploy this model if the validation-test gap widens or if the model's performance on the minority class degrades significantly.

11 · Artifact manifest (Table 14)

artifact	size	description
tfidf_vectorizer.joblib	329 KB	TF-IDF featurizer (30k, 1-2 gram, sublinear, min_df
model_logistic_regression.joblib	4 KB	fitted candidate (CHAMPION)
model_xgboost.joblib	125 KB	fitted candidate (challenger)
model_lightgbm.joblib	1,326 KB	fitted candidate (challenger)
split_indices.json	41 KB	exact train/validation/test membership (reproducib
environment.txt	0 KB	package versions used for this run
(bert_finetuned weights)	—	fine-tuned distilbert-base-uncased (epochs=2, max

11 · Environment (Table 15)

package	version
python	3.11.15
numpy	1.26.4
pandas	2.2.3
scikit-learn	1.5.2
xgboost	2.1.1
lightgbm	4.7.0
torch	2.13.0
transformers	5.14.1
matplotlib	3.9.2

Sign-off

Prepared under the bank's model risk management framework (SR 11-7 / OCC 2011-12).

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Independent review: submitted to second-line validation with this document, the data profile, and the artifact set under `docs/model\_development/artifacts/`.